MATH FOUNDATIONS

Discrete Math for Computer Science

Sets, logic, induction, counting, and the pigeonhole principle — the math underneath every algorithm.

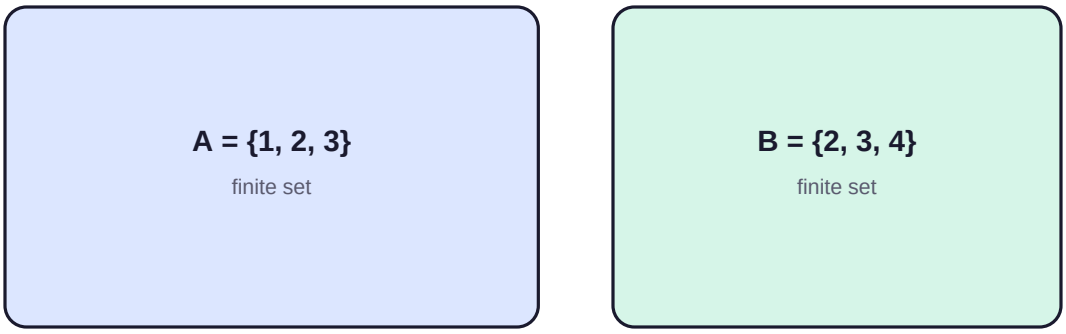
ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

Sets: The Atoms of CS

A set is an unordered collection of distinct objects. Everything starts here.



A intersect B = {2, 3}

A union B = {1, 2, 3, 4}

Key notation: x in A (membership) A subset of B A \ B (difference)

Databases, types, and access-control lists are all set operations in disguise.

STEP 1/6

Logic: Reasoning Without Ambiguity

Propositions combine with AND, OR, NOT. Truth tables enumerate every case.

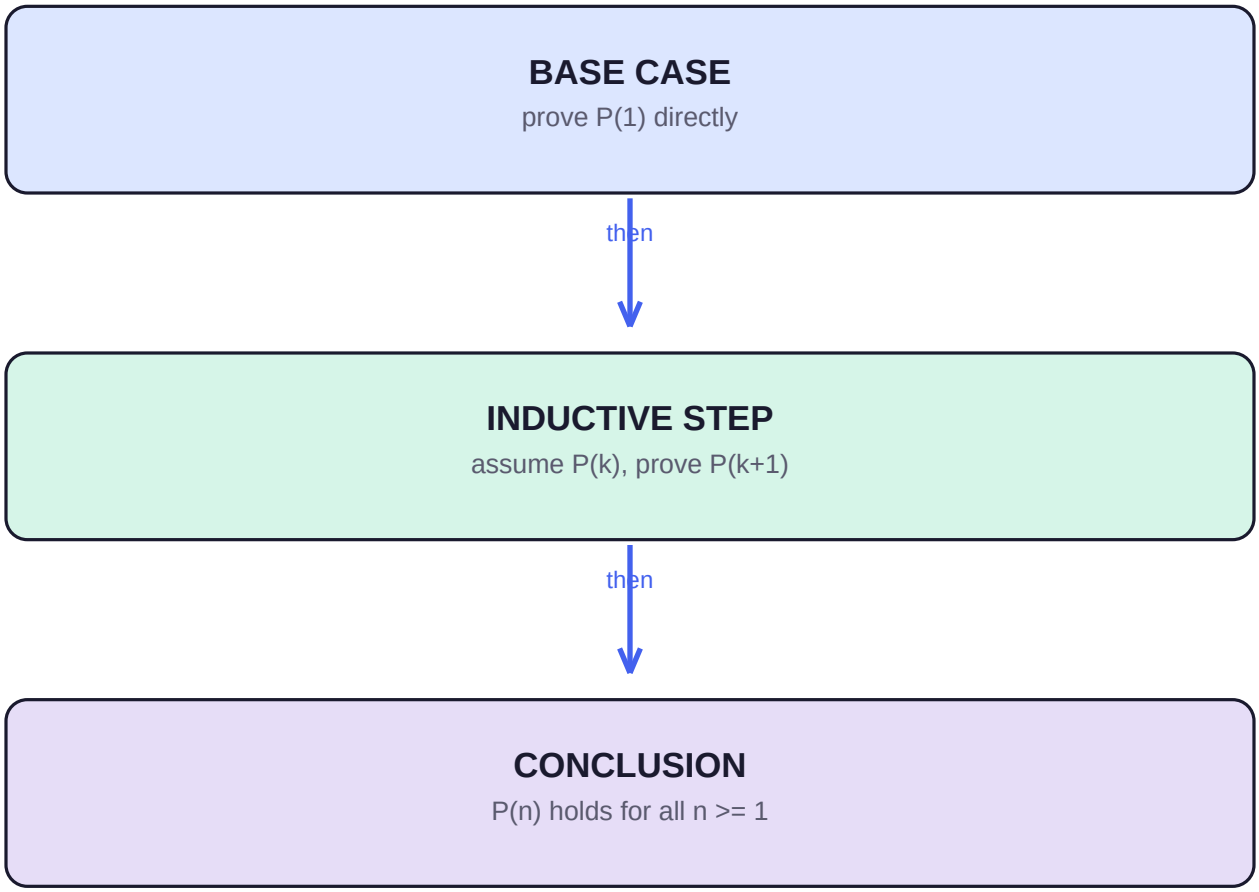
P AND Q	P=T	Q=T	=T
P AND Q	P=T	Q=F	=F
P OR Q	P=F	Q=F	=F
P OR Q	P=T	Q=F	=T

Implication P -> Q is only false when P is true and Q is false — the trap in half of all bug hunts.

STEP 3/6

Proof by Induction

Knock down the first domino, then prove every domino topples the next.



Induction is recursion's mathematical twin — the proof structure of every recursive algorithm's correctness.

STEP 4/6

Combinatorics: Counting Without Listing

Two rules answer nearly every counting question.

PERMUTATIONS

order matters: n! /(n-k)!

COMBINATIONS

order free: n! / k!(n-k)!

MULTIPLICATION RULE

k independent choices with n1..nk options -> n1 * n2 * ... * nk

ADDITION RULE

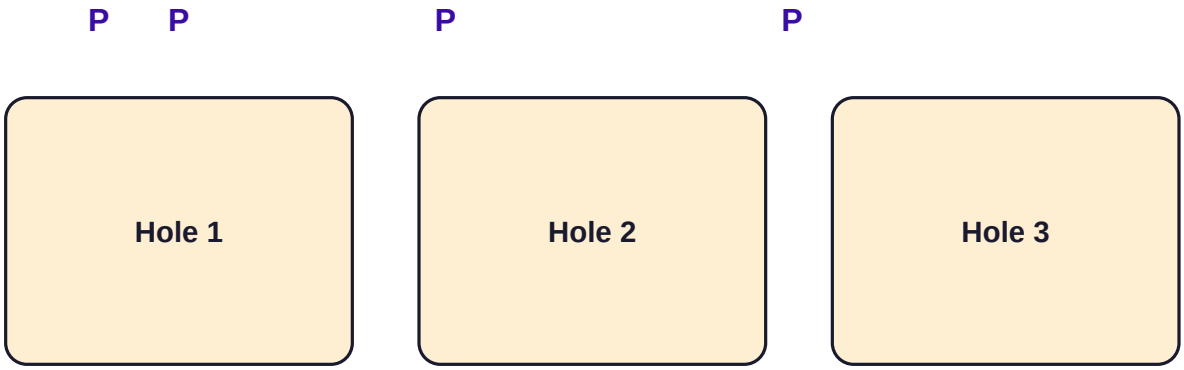
mutually exclusive cases -> add their counts

Passwords, shuffles, committee picks, packet routes: choose the right rule and the number falls out.

STEP 5/6

The Pigeonhole Principle

n items, m containers, n > m -> at least one container holds 2+ items.



4 pigeons, 3 holes -> Hole 1 must hold 2 pigeons. No escape.

POWER USE

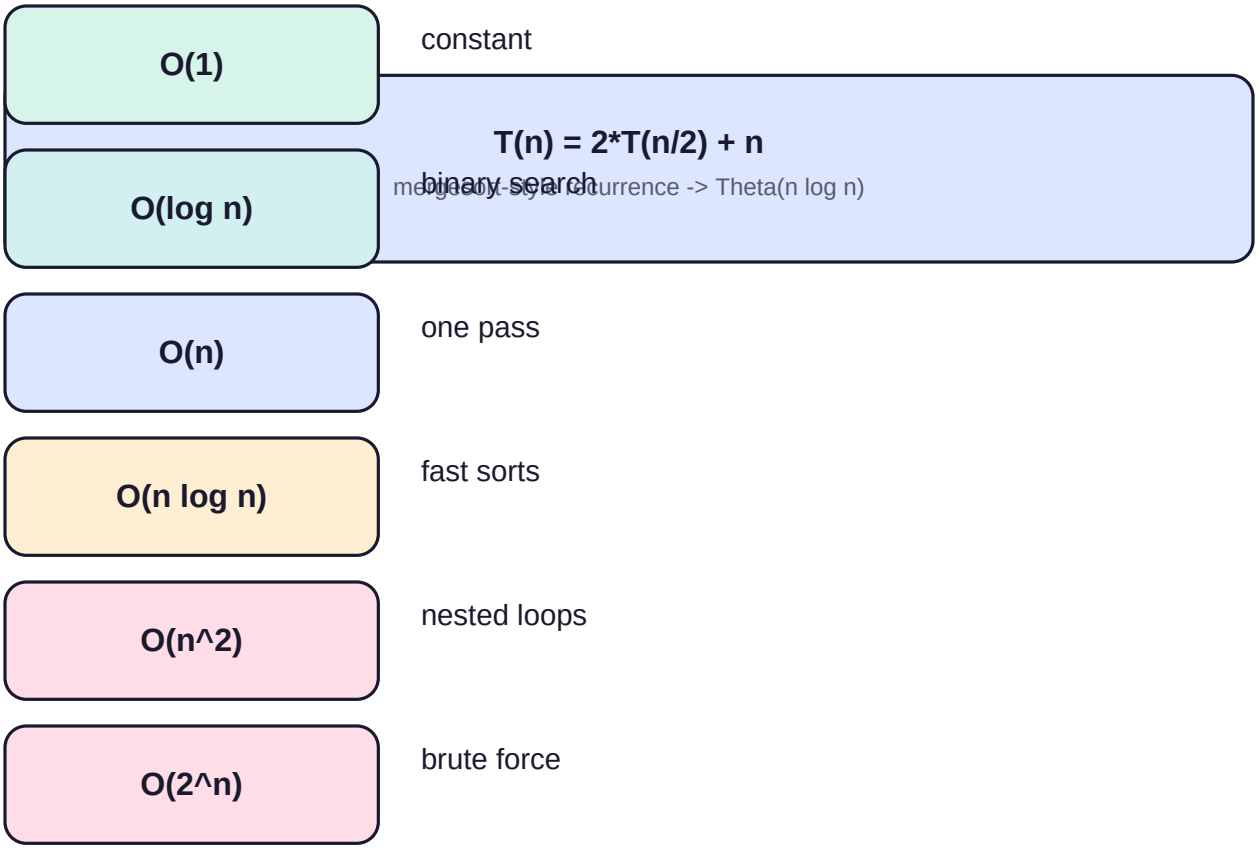
hash collisions MUST exist: 2^256 inputs, 2^256 outputs — wait, that matches...

General form: n items into m holes -> one hole has >= ceil(n/m). Collisions, birthdays, compression limits all follow.

STEP 6/6

Growth Orders: What Induction + Counting Buy You

Recurrences describe algorithms; growth orders compare them.



STEP ✓I✓

Key Takeaways

Discrete Math — Remember

- Sets and their operations (union, intersection, difference) model data collections everywhere.
- Logic and truth tables give unambiguous rules for reasoning about program correctness.
- Induction proves statements for all n: base case plus the k -> k+1 step.
- Permutations count ordered selections; combinations count unordered ones; multiply or add accordingly.
- The pigeonhole principle guarantees collisions whenever items outnumber containers.
- Recurrences plus growth orders let you compare algorithms before writing a line of code.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev